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Cybersecurity and the CIA Triad

1. Introduction

Cybersecurity is the practice of protecting computers, networks, applications and data from unauthorized access, attacks, damage and disruption. It involves technical controls, policies, user awareness and continuous monitoring to keep digital systems secure.

Cybersecurity is important to individuals, businesses, schools, banks and government institutions because they depend on digital systems to store information and deliver services.

2. The CIA Triad

The CIA triad is a fundamental cybersecurity model consisting of Confidentiality, Integrity and Availability.

Principle	Meaning	Real-world example	Security measures
Confidentiality	Only authorized people should access information.	A bank customer can view their own account using authenticated access.	Access controls, encryption, MFA
Integrity	Information must remain accurate and protected against unauthorized changes.	A school prevents unauthorized changes to students' examination results.	Hashing, audit logs, digital signatures, permissions
Availability	Systems and information should be accessible when authorized users need them.	A hospital maintains backups and redundant systems so staff can access patient records during an outage.	Backups, redundancy, monitoring, disaster recovery

3. Why the CIA Triad Matters

- Confidentiality protects privacy and sensitive information.
- Integrity helps users trust that data is accurate and complete.
- Availability helps organizations maintain essential services.
- A security incident may affect one or more of these principles.

4. Conclusion

The CIA triad provides a simple framework for identifying cybersecurity objectives. Security teams should apply appropriate controls to protect confidentiality, maintain integrity and ensure availability.